

SUMMARY

This video explains the mathematical structure of equivalence relations, partitions, Bell numbers, and partial order relations. It establishes the formal one-to-one correspondence between equivalence relations and partitions, explains how Bell numbers quantify such relations, and transitions into partially ordered sets (Posets) and totally ordered sets (Totsets).

Critical insights include:

One-to-one correspondence between equivalence relations and partitions.
Bell numbers as the count of equivalence relations on a set of size n .
Ordered pair count formula derived from partition block sizes.
Structural contrast between equivalence relations and partial orders.
Comparability as the defining feature of totally ordered sets.

I. EQUIVALENCE RELATIONS AND SET PARTITIONS

An equivalence relation on a set A satisfies:

Reflexive
Symmetric
Transitive

1.1 Key Definitions

Equivalence Class

For $x \in A$:

$$[x] = \{ y \in A \mid (x, y) \in R \}$$

Partition of a Set

A partition of A is a collection of non-empty subsets $\{A_1, A_2, \dots, A_k\}$ such that:

Condition	Requirement
Disjoint	$A_i \cap A_j = \emptyset$ for $i \neq j$
Exhaustive	$A_1 \cup A_2 \cup \dots \cup A_k = A$
Non-empty	Each $A_i \neq \emptyset$

1.2 Fundamental Correspondence

From Relation to Partition

Distinct equivalence classes form a partition of A.

From Partition to Relation

Given partition $\{A_i\}$:

$$R = \bigcup (A_i \times A_i)$$

Conclusion

Number of equivalence relations on A
= Number of partitions of A

II. BELL NUMBERS

The number of equivalence relations on a set of size n equals the nth Bell number B_n .

2.1 Bell Numbers Table

n	B_n
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0	1
---	---

1	1
---	---

2	2
---	---

3	5
---	---

4	15
---	----

5	52
---	----

6	203
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2.2 Bell Triangle Construction

Step 1

Start with $B_0 = 1$.

Step 2

Each new row begins with the last element of the previous row.

Step 3

Each subsequent entry equals the sum of:

Current entry + entry above it.

Step 4

The last number in the nth row is B_n .

III. CARDINALITY OF EQUIVALENCE RELATIONS

If a partition of A has block sizes:

$s_1, s_2, \dots, s_k$

Then the number of ordered pairs in the corresponding equivalence relation is:

$$\text{Total Pairs} = s_1^2 + s_2^2 + \dots + s_k^2$$

3.1 Case Study: $|A| = 6$ with 12 Ordered Pairs

We seek partitions of 6 such that:

Sum of squares = 12

Valid partitions:

Combination 1

Sizes: 1, 1, 1, 3

$$1^2 + 1^2 + 1^2 + 3^2 = 12$$

Number of relations:

$$(6C1 \times 5C1 \times 4C1 \times 3C3) / 3! = 20$$

Combination 2

Sizes: 2, 2, 2

$$2^2 + 2^2 + 2^2 = 12$$

Number of relations:

$$(6C2 \times 4C2 \times 2C2) / 3! = 15$$

Total relations:

$$20 + 15 = 35$$

IV. PARTIAL ORDER RELATIONS AND POSETS

A partial order relation satisfies:

Reflexive

Antisymmetric

Transitive

4.1 Poset Definition

A Poset is denoted:

(A, R)

Where:

A = Set

R = Partial order relation

4.2 Examples of Posets

Structure	Relation
$(\mathbb{R}, \leq)$	Usual numerical order
$(\mathbb{Z}^+,)$	
$(P(S), \subseteq)$	Subset relation

4.3 Structural Distinctions

Symmetric vs Antisymmetric

If non-diagonal pairs exist, a relation cannot be both symmetric and antisymmetric.

Diagonal Relation (ΔA)

$$\Delta A = \{(a, a) \mid a \in A\}$$

This is the only relation that is both:

Equivalence relation

Partial order relation

Domain Constraint Example

Divisibility is a partial order on positive integers.

It fails on all integers including zero due to reflexivity issues.

V. COMPARABILITY AND TOTALLY ORDERED SETS

5.1 Comparability

In a Poset (A, R) :

Elements a and b are comparable if:

$$(a, b) \in R \text{ or } (b, a) \in R$$

Reflexivity ensures each element is comparable with itself.

In many Posets, some pairs remain incomparable.

Example

In divisibility on $\{1, 2, 3, 4\}$:

2 and 3 are incomparable.

5.2 Totally Ordered Set (Toset)

A Toset (Chain) is a Poset where:

Every pair of elements is comparable.

Property

Every Toset is a Poset.

Not every Poset is a Toset.

Examples

Structure	Type
$(\mathbb{R}, \leq)$	Always a Toset
Divisibility	Poset, not always Toset
Subset relation	Poset, not always Toset

Conditional Totality

Subset or divisibility relations become Tosets only when restricted to specific chains.

VI. FINAL SYNTHESIS

Equivalence relations partition sets into disjoint equivalence classes.

Bell numbers count these partitions.

Ordered pair count equals sum of squares of block sizes.

Partial orders replace symmetry with antisymmetry.

Comparability distinguishes general Posets from Totally Ordered Sets.

This completes the structural bridge from equivalence relations to ordered systems in discrete mathematics.

